Consciousness and Cognition 120 (2024) 103682

Contents lists available at ScienceDirect

Consciousness and Cognition

journal homepage: www.elsevier.com/locate/yccog

# Task relevance alters the effect of emotion on congruency judgments during action understanding☆

Yiheng Chen [a], Qiwei Zhao [a], Yueyi Ding [a], Yingzhi Lu [a,b,c,*]

[a] School of Psychology, Shanghai University of Sport, China
[b] Research Center for Exercise and Brain Science, Shanghai University of Sport, China
[c] Key Laboratory of Sports Cognition Assessment and Regulation of the General Administration of Sport of China, Shanghai University of Sport, China

ARTICLE INFO

*Keywords:*
Task relevance
Congruency judgment
Action understanding
Affective stimulus–response compatibility effect
Attention

ABSTRACT

The congruency judgments in action understanding helps individuals make timely adjustments to unexpected occurrence, and this process may be influenced by emotion. Previous research has showed contradictory effect of emotion on conflict processing, possibly due to the degree of relevance between emotion and task. However, to date, no study has systematically manipulated the relevance to explore how emotion affects congruency judgments in action understanding. We employed a cue-target paradigm and controlled the way emotional stimuli were presented on the target interface, setting up three experiments: emotion served as task-irrelevant distractor, task-irrelevant target and task-relevant target. The results showed that when emotion was irrelevant to the task, it impaired congruency judgements performance, regardless of a distractor or a target, while task-relevant emotion facilitated this process. These findings indicate that the impact of emotion on congruency judgements during action understanding depends on the degree of emotion-task relevance.

## 1. Introduction

The processing of emotional information is crucial for the adaption of behavior to ever-changing, highly volatile environments, and the ability of an emotional stimulus to capture attention influences cognitive control over goal-directed behaviors (Dolcos & McCarthy, 2006). For instance, in the context of dynamic motor interactions, individual behavior reflects complex cognitive processes, including the encoding of others' actions, the anticipation of outcomes (Moreau et al., 2020, 2022), the assessment of congruency between anticipated and actual outcomes (Zhao et al., 2021), and the resolution of any conflict encountered (Moreau et al., 2020, 2022). Conflict resolution has been investigated in numerous studies using cognitive control paradigms such as the Stroop, Simon, and flanker tasks, the performance in which has been shown to be influenced by emotional stimuli (Roh et al., 2016; Xue et al., 2013; Yuan et al., 2011). However, these studies focused primarily on the role of emotions in conflict resolution, that is, implicit conflict processing (as in Stroop task, where is no need to judge the presence of conflict, but merely to respond to the word meaning or color). On the other hand, explicit conflict processing, such as congruency judgments, remains underexplored in terms of the impact of emotions on this process.

☆ We describe the determination of sample size, data exclusion criteria, and all analysis methods employed in this study. The data, processing code, stimulus materials and experimental procedures are publicly available at https://osf.io/r6vqz/. This study's design and its analysis were not preregistered.

* Corresponding author at: School of Psychology, Shanghai University of Sport, 650 Qing Yuan Huan Road, Shanghai 200438, China.
*E-mail address:* luyingzhi@sus.edu.cn (Y. Lu).

https://doi.org/10.1016/j.concog.2024.103682
Received 9 July 2023; Received in revised form 1 March 2024; Accepted 19 March 2024
Available online 29 March 2024
1053-8100/

A table tennis serve exemplifies congruency judgements within the context of action understanding. During such a serve, individuals anticipate the ball's landing point using the server's kinematic information and the ball's partial trajectory (Zhao et al., 2018). When the actual outcome occurs, its deviation from the anticipated outcome can be determined and used to judge whether incongruence exists for further cognitive processing (Zhao et al., 2021). The Predicted Response Outcome (PRO) model provides a theoretical basis for such congruency judgement to monitor for unexpected occurrences (Alexander & Brown, 2011). The detection of incongruency induces neural activity in the anterior cingulate cortex (ACC) to deal with the conflict (Alexander & Brown, 2017; Brown, 2013). Such judgement occurs not only in behavior monitoring, but also in action understanding, as demonstrated by the detection of significant theta oscillation activation in the middle frontal gyrus during unexpected occurrences (Lu et al., 2020; Zhao et al., 2021). Moreover, emotional processing also activates the ACC (Bush et al., 2000; Etkin et al., 2011), and conflict processing in the presence of incongruency is automatically evaluated as aversive emotional signaling (Dignath et al., 2020; Dreisbach & Fischer, 2015; Inzlicht et al., 2015). In other words, negative affect and conflict processing have overlapping neural representation (Braem et al., 2017; Shackman et al., 2011; Vermeylen et al., 2020), and considering that congruency judgments inherently embody conflict processing, it is plausible to speculate that emotion and congruency judgement during action understanding may interact with each other.

The interaction between emotion and congruency judgments may reveal the affective stimulus–response compatibility (SRC) effect. This effect has been observed in studies of the effects of emotion on approach–avoidance behaviors. Humans instinctively approach positive and avoid negative stimuli (Chen, 1999). A compatible effect manifests in tasks that align with this inclination (positive approaching and negative avoiding), whereas an incompatible effect arises in tasks that contradict it (positive avoiding) (Eder & Rothermund, 2008; Kozlik et al., 2015; Krieglmeyer & Deutsch, 2010; Xia et al., 2021). The Theory of Event Coding provides an explanation for the affective SRC effect, positing the existence of a feature-based coding system for perceived and to-be produced events and thereby integrating stimuli and responses into a common framework (Hommel et al., 2001; Lavender & Hommel, 2007). This integration suggests that a compatible effect arises when stimulus and response set features overlap, whereas an incompatible effect occurs in the absence of such overlap. Similarly, the features of congruency response sets may overlap with those of valence-based stimulus sets. As previously mentioned, congruency judgments inherently embody conflict processing. According to the PRO model, unexpected occurrences are evaluated as a negative surprise (Brown, 2013). In other words, after evaluative response coding, the response labels in congruency judgements share valence dimensions with emotional stimuli. Hence, we can infer that emotions affect congruency judgements in action understanding, manifesting the affective SRC effect.

A wide range of work has explored the roles of emotions on cognition and behavior and their influences thereon, but the findings have been inconclusive, suggesting that the influence of emotional valence on cognition is modulated by factors such as attention. Negative emotional distractors have been shown to impair working-memory congruency judgments (Dolcos & McCarthy, 2006) and conflict resolution in the Stroop task (Hart et al., 2010), but opposite effects have been observed in other studies. For example, negative target stimuli were found to enhance conflict resolution in the flanker (Kanske & Kotz, 2011b) and Simon tasks (Kanske & Kotz, 2011a). We speculate that the manner in which emotions are evoked in the tasks may lead to different outcomes. Notably, emotions as distractors or targets did not require valence processing to accomplish the primary tasks in those studies. Dou et al. (2021) found more rapid identification of negative than neutral facial targets in tasks requiring valence recognition (in which emotion is task relevant), but not those in which emotion was not task relevant. Thus, in addition to differences in experimental paradigms, the inconsistency of previous results may be due to the roles and degrees of task relevance of emotion.

The effects of emotions have been found to be contingent on their task relevance in studies of approach–avoidance behavior. Lavender and Hommel (2007) conducted a study in which participants made approach and avoidance responses based on the valence or orientation of emotionally charged pictures. The researcher observed the affective SRC effect only in the valence task, supporting the conclusion that valence-prompted response initiation tendencies are not adequately automatic to operate without conscious affective evaluation (Klauer & Musch, 2002; Rotteveel & Phaf, 2004). Appraisal theories of emotion offer theoretical support for this finding, proposing that emotional stimuli do not automatically trigger approach–avoidance behaviors, but that such triggering depends on the task relevance of the emotion (Moors & Fischer, 2019; Scherer & Moors, 2019). In other words, emotions' effects on behavior differ according to their context-specific relevance. Recent research further corroborates this view; Welsch (2023) found that participants' approach and avoidance responses were based on the valence or gender of facial stimuli, and that the affective SRC effect manifested only after conscious affective evaluation during valence task performance. Moreover, research has revealed that only task-relevant emotional stimuli affect behavioral performance in other task types (Dou et al., 2021; Mancini et al., 2022; Mirabella, 2018; Mirabella et al., 2023). Overall, these findings suggest that task relevance alters the effects of emotion on congruency judgments during action understanding, and that the affective SRC effect manifests in congruency judgments only when emotions are task relevant.

The present study was conducted to examine whether task relevance altered the impact of emotion on congruency judgments during action understanding using a variant cue-target task in an action understanding framework (Lu et al., 2020). We hypothesized that the impact of emotions on congruency judgments would (1) reflect an affective SRC effect and (2) vary with task relevance. Participants watched a video of an incomplete table-tennis serve and anticipated the ball landing outcome based on the server's kinematic cues and the partial ball's trajectory in the air. When the actual outcome appeared on the target interface, the participants judged its congruency with the anticipated outcome. To systematically vary the degree of task relevance, we manipulated the presentation of emotional stimuli on the target interface as (1) task-irrelevant distractors, (2) task-irrelevant targets, and (3) task-relevant targets.

## 2. Experiment 1: Task-Irrelevant distractor

The aim of experiment 1 was to explore the effects of task-irrelevant emotional distractors on congruency judgments during action understanding. For that purpose, participants performed a cue-target congruency judgement task. To manipulate the relevance between emotion and the task, both a target and an emotional distractor were presented in the target display. On the basis of previous research findings regarding the impairment effect of emotional distractors (Dolcos & McCarthy, 2006; Hart et al., 2010), we predicted that the emotional distractor would have an inhibitory effect on the performance of the main task. Additionally, as the manifestation of affective SRC effect might depend on the relevance between emotions and the task (Mirabella et al., 2023; Welsch et al., 2023), we predict a reduced affective SRC effect under task-irrelevant.

### 2.1. Methods

#### 2.1.1. Transparency and data accessibility statement

All reported studies followed APA's ethical principles for research with human participants during data collection. All measures, manipulations, and exclusions in the studies are disclosed, and hypothesis test analysis started only after data collection was finished. The data, processing code, stimulus materials and experimental procedures are publicly available at OSF https://osf.io/r6vqz/. The data were analyzed using R 4.2 and SPSS 26.0. This study's design and its analysis were not preregistered.

#### 2.1.2. Participants

Thirty-eight subjects (24 females; $M_{age} = 21.316$ years, $SD = 0.440$) were obtained after passing pre-experiment, screening test. All participants had no prior experience with table tennis. Sensitivity analysis using G*Power 3.1 (Faul et al., 2007) for the two-factor within-subjects repeated measures ANOVA indicates that the sample of 38 participants is sufficient to detect even small effect size ($f = 0.193$, $\eta 2\, p = 0.036$) with $\alpha = 0.05$ and power = 0.90. The effects of interest in this analysis are the main effects of each factor and their interaction. All subjects were not informed about the purpose of the study and agreed to participate and none of them participated in similar experiments. All participants possessed normal or corrected-to-normal vision, no psychiatric history (adopted *State-Trait Anxiety Inventory*, $M = 31.853$, $SD = 1.277$) and right-handed (adopted *Edinburgh Handedness Inventory*, $M = 86.571$, $SD = 2.926$). Participants were compensated 60 CNY per hour for the experiments. This work was approved by the local ethics committee (No. 102772020RT111). The data of three experiments was collected in 2021.

#### 2.1.3. Stimuli and apparatus

Stimuli were presented with E-prime 2.0 on a 15-inch color monitor. Participants sat in a distance of 50 cm from the monitor in a bright experimental cabin.

Twenty videos depicting a female table tennis player serving, with an equal probability of serving to the left and right, were recorded from the perspective of her opponent (Canon 5D Mark III; resolution, 1,280 × 720 pixels at 60 frames per second, fps). Previous research has demonstrated that individuals with limited experience in ping pong who viewed videos that ended at a frame after the racket-ball contact were able to effectively predict the landing point of the ball (Lu et al., 2020). The selected videos underwent the following processing using Adobe Premiere software (Adobe Systems Incorporated; San Jose, CA, USA): (a) the audio tracks were muted, (b) the frame size and rate were changed to a video resolution of 640 × 360 pixels at 50 fps (i.e., 20 ms per frame) to provide the best viewing experience for participants, (c) the video ended one frame after racket-ball contact, was then rewound 50 frames to the start of the video, resulting in a 1,000 ms (50 frames), and (d) the server's face in each video was blurred to eliminate the impact of facial features and head motion.

The emotional pictures were selected from the International Affective Picture System (IAPS) and consisted of 40 positive, 40 negative, and 40 neutral images.[1] An independent samples *t*-test showed that the valence of positive images ($M = 7.415$, $SD = 0.546$) was significantly higher than that of negative ($M = 2.718$, $SD = 0.945$) and neutral images ($M = 4.900$, $SD = 0.193$). There was no significant difference between the arousal level of positive ($M = 5.337$, $SD = 0.934$) and negative images ($M = 5.706$, $SD = 0.798$), both of which were significantly higher than that of neutral images ($M = 2.915$, $SD = 0.647$) (Table 1). To minimize participant processing of biological movement information, the images were selected without any human-related information. The size of the images was 480 × 360 pixels.

The stimulus representing the outcome landing point was a ping-pong ball, which could locate on the left or right side of the screen. The size of the ball was 60 × 60 pixels.

[1] The positive pictures are assigned the following numbers: 1710, 1750, 1460, 1440, 8501, 5910, 7502, 7330, 8170, 8496, 8210, 7270, 5480, 8502, 7230, 8030, 8185, 5621, 5629, 8190, 5760, 5830, 1920, 1610, 1463, 8500, 1600, 5849, 7402, 7400, 7260, 7460, 7289, 8503, 7220, 7570, 7501, 7282, 1540, 5450. The negative pictures are assigned the following numbers: 3053, 3102, 3000, 3064, 3080, 3015, 3168, 3130, 3100, 9040, 9570, 3010, 9140, 9181, 3061, 2053, 9430, 9320, 1111, 1270, 9440, 1050, 1274, 1930, 1120, 9561, 9180, 9470, 7361, 7360, 6940, 1030, 9490, 9480, 1220, 9571, 1280, 1240, 6800, 1275. The neutral pictures are assigned the following numbers: 7920, 7031, 7595, 7025, 7186, 7030, 7040, 7150, 7180, 7590, 7130, 7705, 5535, 7211, 7217, 7491, 7175, 7006, 7009, 7050, 7010, 7950, 7235, 7002, 7020, 7185, 7035, 7000, 7160, 7004, 7187, 6150, 7233, 7170, 5510, 5531, 7207, 7182, 2880, 7090.

**Table 1**
Comparison of Valence and Arousal of Three Types of Emotional Pictures.

| | Positive | Negative | Neural | Comparisons |
|---|---|---|---|---|
| Valence | $7.415 \pm 0.546$ | $2.718 \pm 0.945$ | $4.900 \pm 0.193$ | P > N; P > Neu; Neu > N; |
| Arousal | $5.337 \pm 0.934$ | $5.706 \pm 0.798$ | $2.915 \pm 0.647$ | P = N; P > Neu; N > Neu; |

#### *2.1.4. Design and procedure*

The experiment comprised two stages, screening test and the main task. The aim of screening test was to filter participants by requiring them to correctly anticipate the landing orientation of the table tennis after viewing a serve clip. Participants were deemed to have adequate anticipation ability if their accuracy surpassed the chance level. Each trial started with a black fixation at the center of the screen for 1,000 ms, followed by the presentation of the 1,000 ms video clip. Subsequently, a 1,500 ms blank screen was presented, during which participants were required to make a judgement about the landing orientation. This was followed by a 500 ms blank screen before proceeding to the next trial. The screening test consisted of 1 block, 40trials, with an equal distribution of left and right landing orientation for the serve clip.

The main task was designed to test the effect of task-irrelevant emotional distractors on congruency judgment during action understanding. We employed a modified cue-target task. At the beginning of each trial, a black fixation point "+" was presented on the center of the screen for 800 ms. Then a cue video depicting a table tennis serve was presented for 1,000 ms, ending with a frame following racket-ball contact, which was intended to instruct the participants to form an expectation regarding location of the ball on the left or right side. After the cue video, a 500 ms blank screen was displayed, followed by the presentation of the target stimulus for 800 ms. The target interface consisted of an emotional picture in the center of the screen and a ball on either the left (represented left landing point) or right (represented right landing point) side of the screen. At this time, the emotion picture was presented as a task-irrelevant distractor. Participants was required to determine whether prior expectation was congruent with the current landing location during target interface. The response time was limited to 2,000 ms. The button assignments for congruent and incongruent responses were either "UPARROW" or "DOWNARROW" and counterbalanced among the participants. To ensure that participants were forming anticipations based on the cue video, probe trials were incorporated into the experiment. In these trials, participants were instructed to answer their prediction regarding the landing point after viewing the serve video. Both the normal trials and probe trials were presented randomly.

The experimental employed a within-subject two-factor design: congruency condition (congruent, incongruent) and emotion valence (negative, neutral, positive). Before the formal experiment, participants completed a practice session of 12 trials to ensure understanding. The formal experiment consisted of 4 identical blocks of 130 trials each (120 normal and 10 probe trials). The landing points of clue videos and consistency between clue videos and target stimuli were equal. One limitation of the design was that each emotional picture appeared once in each block, either in congruent or incongruent trials, but not in both conditions, resulting in 80 trials per condition for each participant.

#### *2.1.5. Statistical analysis*

38 participants were selected, all of whom had an accuracy rate greater than chance level in the screening test. During the main experiment, we assessed each participant's accuracy in the probe trials. Participants' performances were all above chance level, so no exclusions were made. Statistical analysis was performed using SPSS, version 26.0 (IBM SPSS, Inc., Chicago, IL, USA). Only correct trials were selected for the statistical analysis of reaction times, it indicated that the individuals processed the congruency accurately, facilitating observation of emotion's impact on main task performance. For both accuracy and reaction time, the $2 \times 3$ repeated-measures ANOVA with the factors congruency condition (congruent, incongruent) and emotion valence (negative, neutral, positive) was conducted. $p$ value less than 0.050 were considered statistically significant. The sphericity assumption was evaluated using Mauchly's test, and the Greenhouse-Geisser correction for the degrees of freedom was used in case of non-sphericity. The Bonferroni correction was used to correct for multiple post hoc comparisons. The effect size for the statistically significant factors was estimated using partial eta squared ($\eta 2\,p$). We examine the impact emotions on congruency judgements and the their affective SRC effects. As for reaction time analysis, a longer response time for negative emotions compared to neutral under congruent conditions signals an incompatibility effect. Conversely, under incongruent conditions, shorter response times for negative emotions imply a compatibility effect. The simultaneous manifestation of both effects suggests a bilateral symmetric affective SRC effect.

### *2.2. Results*

#### *2.2.1. RT analysis*

The main effect of the congruency between cue and target location was significant, $F\,(1, 37) = 56.362$, $p < .001$, $\eta 2\,p = .604$. RTs in the congruent condition ($M = 723.096$ ms, $SD = 22.465$, 95 % CI [677.579, 768.614]) were significantly shorter than those in the incongruent condition ($M = 780.822$ ms, $SD = 22.676$, 95 % CI [734.876, 826.767]). The main effect of valence was also significant, $F\,(2, 74) = 6.112$, $p = .003$, $\eta 2\,p = .142$. RTs in the negative condition ($M = 762.124$ ms, $SD = 23.449$, 95 % CI [714.612, 809.637]) were significantly longer than those in the neutral condition ($M = 743.594$ ms, $SD = 21.985$, $p = .010$, 95 % CI [699.049, 788.139]), while RTs in the positive condition ($M = 750.159$ ms, $SD = 21.900$, 95 % CI [705.785, 794.533]) did not differ from those in the neutral or negative conditions. Furthermore, the interaction effect between congruency and valence was significant, $F\,(2, 74) = 3.502$, $p = .035$, $\eta 2\,p = .086$. Further simple effect analyses revealed that in the congruent condition, RTs in the negative condition ($M = 740.316$ ms, $SD$

= 24.365, 95 % CI [690.948, 789.683]) were significantly longer than those in the neutral (*M* = 710.382 ms, *SD* = 22.228, *p* =.011, 95 % CI [665.343, 755.421]) and positive (*M* = 718.591 ms, *SD* = 22.137, *p* =.049, 95 % CI [673.738, 763.445]), while RTs in the neutral and positive conditions did not differ. In the incongruent condition, RTs did not differ among the negative, neutral, and positive conditions. These results indicated that when emotions served as task-irrelevant distractors, negative emotions impaired the processing of expected events in congruency judgment, while emotion did not affect this process when unexpected events occurred, specifically revealing an incompatible effect under congruent condition. The findings are illustrated in Fig. 1C.

### 2.2.2. Accuracy analysis

Participants committed more accuracy in congruent (*M* = 0.855, *SD* = 0.011, 95 % CI [0.832, 0.878]) than in the incongruent trials (*M* = 0.813, *SD* = 0.015, 95 % CI [0.783, 0.843]), $F$ (1, 37) = 15.280, $p$ <.001, $\eta 2\ p$ =.292. Nevertheless, there was no main effect or interaction effect of emotion valence (see Fig. 1B).

## 3. Experiment 2: Task-Irrelevant target

The purpose of experiment 2 was to investigate the effect of emotion as a task-irrelevant target on congruency judgments. In the target interface, only one stimulus was presented. That stimulus was an emotional image that was placed on either the left or right side of the screen to show the actual landing location of the ball. Participants then made a congruency judgement between their prediction and the actual outcome. Previous research has shown that emotions as targets facilitate conflict resolution in tasks like the flanker (Kanske & Kotz, 2011b) and Simon (Kanske & Kotz, 2011a). However, studies that contradict this find that task-irrelevant targets do not impact behavior (Mirabella et al., 2023). Thus, predicting the specific effect of emotions in this context is challenging. Nevertheless, we anticipate that the effects produced by distractors and targets will differ.

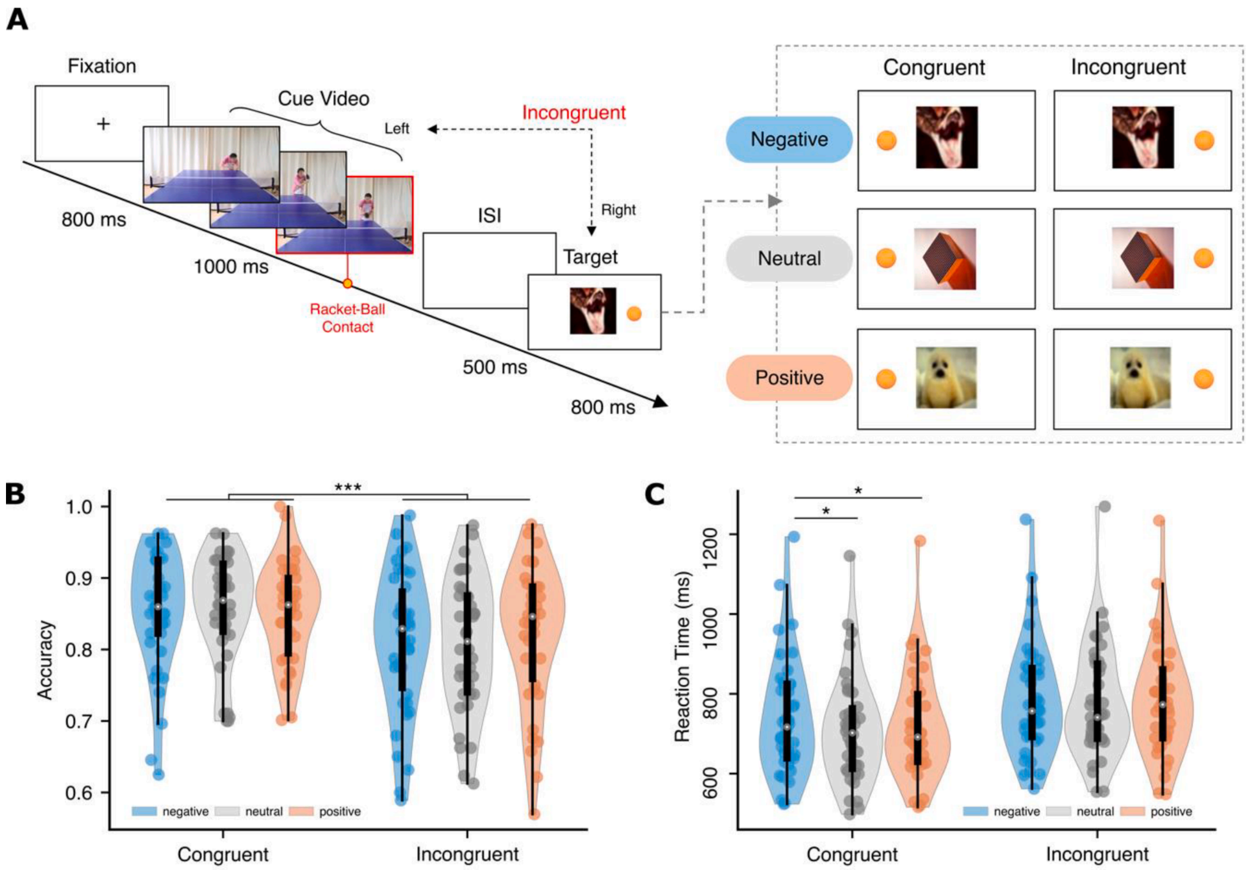


**Fig. 1.** Experiment 1 Procedure and Results. *Note.* (A) Experiment 1 procedure. Each trial started with an 800 ms fixation, followed by a 1,000 ms serving video clip until racket-ball contact. Participants were required to predict the landing point based on the clip. In the subsequent target interface, the actual landing point presented on one side of the screen may be congruent or incongruent with the landing point indicated by the clip, while an emotional distractor was presented in the center. The six experimental conditions were outlined by the dashed box. (B) Accuracy. Violin plots show a significant difference in congruency condition, with points and distributions for data aggregated by subject. (C) Reaction time. The interaction is reflected in the difference between negative and neutral under congruent condition, but no difference is found under incongruent condition. *$p$ <.050; ***$p$ <.001.

### 3.1. Methods

#### 3.1.1. Participants

Thirty-three participants (22 females; $M_{age} = 21.273$ years, $SD = 2.719$) were recruited and passed the screening test. All participants had no prior experience with table tennis. Sensitivity analysis for the two-factor within-subjects repeated measures ANOVA indicates that the sample of 33 participants is sufficient to detect even small effect size ($f = 0.207$, $\eta 2\ p = 0.041$) with $\alpha = 0.05$ and power = 0.90. The effects of interest in this analysis are the main effects of each factor and their interaction. All participants possessed normal or corrected-to-normal vision, no psychiatric history (adopted *State-Trait Anxiety Inventory*, $M = 31.394$, $SD = 7.013$) and right-handed (adopted *Edinburgh Handedness Inventory*, $M = 86.333$, $SD = 16.914$).

#### 3.1.2. Stimuli and apparatus

As in Experiment 1, participants sat in an experimental cabin in a distance of 50 cm from a 15-inch color monitor on which stimuli were presented with E-prime 2.0.

The stimulus consisted of cue videos (same parameters as in Experiment 1) and target stimuli (only emotional images). The emotional pictures were the same as those selected in Experiment 1, and the differences in valence and arousal were detailed in Table 1.

#### 3.1.3. Design and procedure

In contrast to Experiment 1, Experiment 2 featured a target interface with only one stimulus, the emotional picture, which appeared on the screen to indicate the position of the outcome. On the other hand, Experiment 1 had two stimuli, the emotional picture, and the ball (see Fig. 2A). This design aimed to examine whether degrees of relevance that emotion played in overall task could influence affective SRC effect.

At the beginning of each trial, a black fixation was presented in the center of the screen for 800 ms, followed by a 1,000 ms cue video, which instructed participants to form an expectancy of where the ball would land. Then a blank screen was presented for 500 ms, followed by the target interface for 800 ms, which displayed an emotional picture. The picture was located on the left or the right of the screen (indicating the outcome). Participants were required to make a key response in the target interface, judging whether the

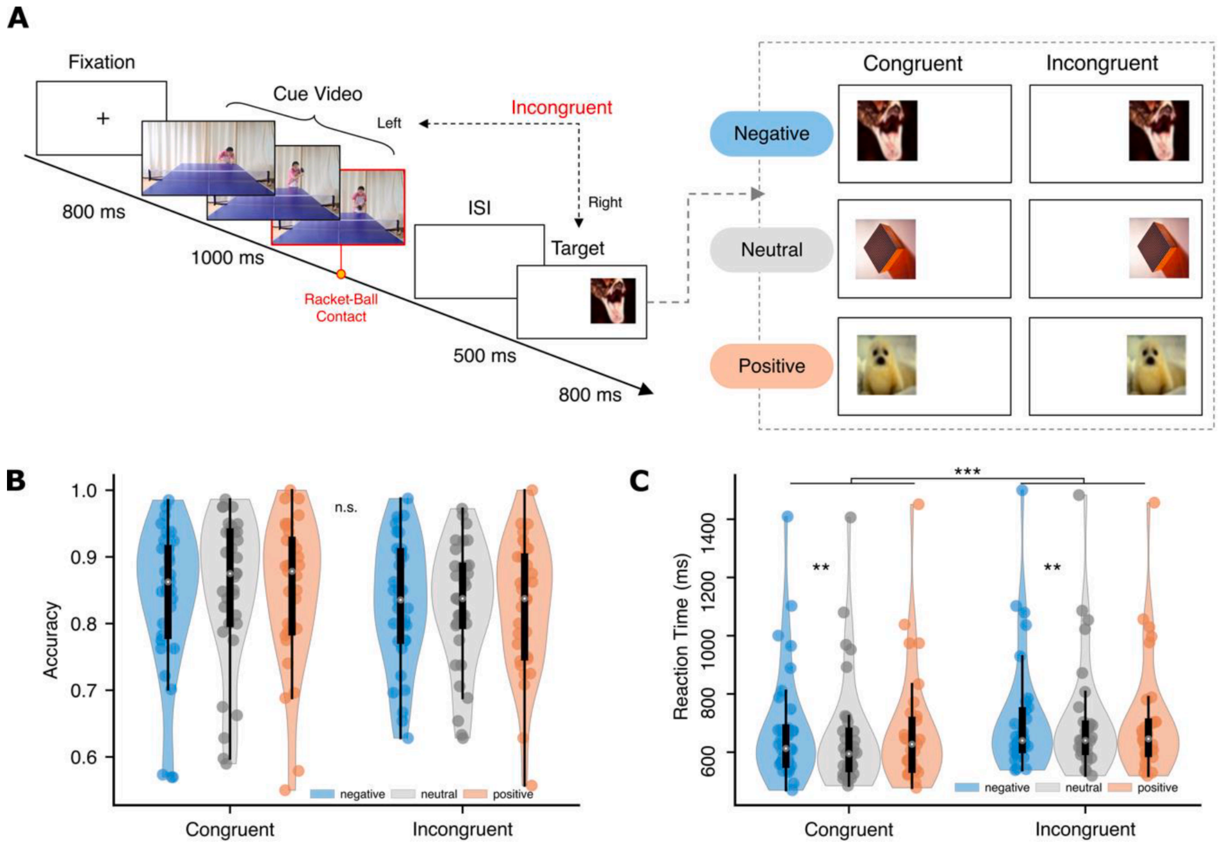


**Fig. 2.** Experiment 2 Procedure and Results. *Note.* (A) Experiment 2 procedure. Unlike experiment 1, in the target interface, there is no ball. Instead, an emotional picture appears on one side of the screen to indicate the actual landing point. In this case, emotion severs as task-irrelevant target. (B) Accuracy. No effect is found. (C) Reaction time. Only a significant effect in congruency is found. **$p$ <.010; ***$p$ <.001.

landing location was congruent with their expectancy or not, with a maximum keypress time of 2,000 ms. Experiment 2 also included probe trials, and all key mappings were the same as in Experiment 1. The experimental conditions and number of trials performed for each condition were identical to those of Experiment 1.

#### 3.1.4. Statistical analysis

After checking the accuracy of each participant's responses in the probe trials, those whose accuracy was lower than the chance level was excluded from the data analysis because their action anticipation was considered unreliable and congruency processing was questionable. All participants' accuracy exceeded 50 %. Statistical tests were computed in the same way as in Experiment 1.

### 3.2. Results

We analyzed RTs with a repeated-measures ANOVA including the factors congruency condition (congruent, incongruent) and emotion valence (negative, neutral, positive). The results were depicted in Fig. 2C. The analysis showed that the main effect of the congruency between cue and target was significant, $F$ (1, 32) = 25.428, $p$ <.001, $\eta 2\, p$ =.443, with significantly shorter RTs in the congruent condition ($M$ = 669.291 ms, $SD$ = 34.220, 95 % CI [599.586, 738.995]) compared to the incongruent condition ($M$ = 707.867 ms, $SD$ = 34.488, 95 % CI [637.618, 778.116], $p$ <.001). The main effect of valence was also significant, $F$ (2, 64) = 7.773, $p$ =.001, $\eta 2\, p$ =.195, with longer RTs for negative ($M$ = 697.529 ms, $SD$ = 34.600, 95 % CI [627.052, 768.007]) compared to neutral ($M$ = 680.941 ms, $SD$ = 34.171, $p$ =.002, 95 % CI [611.338, 750.545]), while there was no significant difference between positive ($M$ = 687.266 ms, $SD$ = 33.911, 95 % CI [618.191, 756.341]) and neutral. The interaction between congruency and valence was not significant, $F$ (2, 64) = 1.518, $p$ =.227.

We analyzed accuracy rate by a repeated-measures ANOVA that included the same factors as the RT analysis. No main effects of the two factors and no interaction effects were found.

## 4. Experiment 3: Task-Relevant target

For the target interface in experiment 3, emotional images were presented as target stimuli. Participants were instructed to identify the valence of the stimulus in order to obtain information about the actual outcome, which they then compared with the cue to make a congruency judgement. Consequently, a conscious affective evaluation was necessary to complete the main task, making emotion relevant to the task. On the basis of previous research findings (Dou et al., 2021; Pessoa et al., 2002), we hypothesized that task-relevant emotional targets would enhance congruency judgments performance. Additionally, we predict a fully affective SRC effect under task-relevant.

### 4.1. Methods

#### 4.1.1. Participants

Twenty-nine participants (19 females, $M$ = 21.200, $SD$ = 2.809) were recruited to take part in the study. All participants had no prior experience with table tennis. Sensitivity analysis for the two-factor within-subjects repeated measures ANOVA indicates that the sample of 29 participants is sufficient to detect even small effect size ($f$ = 0.253, $\eta 2\, p$ = 0.060) with $\alpha$ = 0.05 and power = 0.90. The effects of interest in this analysis are the main effects of each factor and their interaction. All participants were required to pass screening test and had normal or corrected-to-normal vision, were not informed about the purpose of the experiment and were compensated.

#### 4.1.2. Stimuli and apparatus

The stimuli consisted of cue video and target stimuli (only emotional pictures). Since Experiment 1 and 2 showed that only negative and neural emotions had significant differences in RTs, while positive emotions did not differ significantly from neutral. Consequently, Experiment 3 selected 40 negative and 40 neutral pictures. Compared to neutral images, negative exhibited significant variations in both valence and arousal, as indicated in Table 1.

#### 4.1.3. Design and procedure

In contrast to Experiment 2, the emotional images in the target interface of Experiment 3 were fixed to appear only in the center of the screen, instead of on the left or right side of the screen. Participants were informed beforehand that negative emotions represented the outcome landing on the left, while neutral stimuli represented the outcome on the right. The matching of valence and landing

**Table 2**
Counterbalanced Between Participants.

| Subject | Counterbalance Mapping | Keypress Mapping |
|---|---|---|
| S1-S7 & S29 | Negative-Left, Neutral-Right | UP-Congruent, Down-Incongruent |
| S8-S14 | Negative-Left, Neutral-Right | UP-Incongruent, Down-Congruent |
| S15-S21 | Negative-Right, Neutral-Left | UP-Congruent, Down-Incongruent |
| S22-S28 | Negative-Right, Neutral-Left | UP-Incongruent, Down-Congruent |

outcome, congruency and keypress mapping were all counterbalanced between participants, as shown in Table 2. The purpose of this setup was to investigate how emotions affected congruency judgments when they were presented as task-relevant targets (requiring to identify valence to obtain outcome information).

Every trial started with a black fixation, which was presented in the center of the screen for 800 ms, followed by a 1,000 ms cue video, instructing the participants to form an expectation in their minds. Then, a 500 ms blank screen was presented, followed by an 800 ms target interface, which only presented an emotional picture in the center of the screen. If a negative picture appeared, it meant the ball landed on the left side; if a neutral picture appeared, it meant that the ball landed on the right side. Participants were required to judge whether the previous anticipation and the current outcome were congruent, and the keypress duration should not exceed 2,000 ms. The probe trials and normal trials were randomly presented with each block (see Fig. 3). Prior to the formal experiment, participants completed 42 practice trials (36 formal and 6 probe trials) to ensure that they learned the association between valence and actual location. Meanwhile, the emotional pictures shown during the practice block were not included in the formal experiment.

We adopted two-factor within-subjects design, congruency condition (congruent, incongruent) and emotional valence (negative, neutral). Experiment 3 consisted of 6 blocks of 86 trials each (80 normal and 6 probe trials). The expected landing position allocation for cue videos was equal, and the ratio of cue-target congruency was set to 1:1. Each emotional picture appeared once in each block,

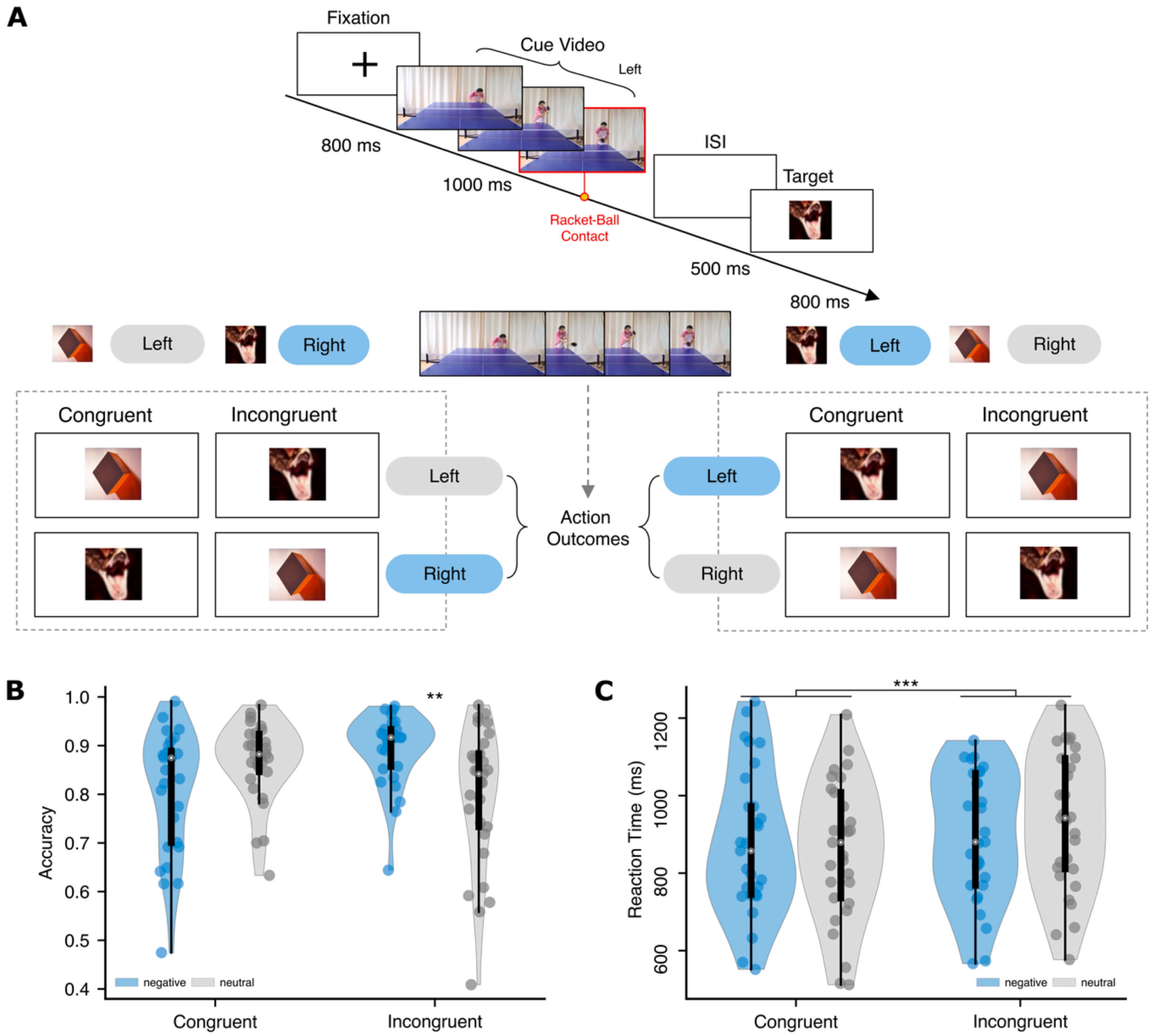


**Fig. 3.** Experiment 3 Procedure and Results. *Note.* (A) Experiment 3 procedure. Upper panel illustrates the procedure (cue video indicates a left landing), while the lower panel depicts two types of valence-location mappings (“negative-left, neutral-right” and “negative-right, neutral-left”). Emotional images are displayed at the center of the screen, and based on the valence-location mapping, participants’ task was to determine whether their prior anticipation align with the actual outcomes. (B) Accuracy. The interaction effect is reflected in that under incongruent condition, negative stimuli induced higher accuracy, while no difference was observed under congruent condition. (C) Reaction time. Only a significant effect in congruency is found. $^{**}p < .010$; $^{***}p < .001$.

resulting in 120 trials per condition for each participant.

#### 4.1.4. Statistical analysis

After checking the accuracy of each participant's responses in the probe trials, those whose accuracy was lower than the chance level was excluded from the data analysis because their action anticipation were considered unreliable and outcome processing was questionable. All participants' accuracy exceeded 50 %. Statistical tests were computed in the same way as in previous experiment.

### 4.2. Results

#### 4.2.1. RT analysis

The results of the two-factor repeated measures ANOVA on RTs were shown in Fig. 3C. The results indicated a significant main effect of congruency between cue and target, $F$ (1, 28) = 19.824, $p$ <.001, $\eta 2\ p$ =.415. Participants responded faster under the congruent condition ($M$ = 868.624 ms, $SD$ = 32.121, 95 % CI [802.827, 934.422]) than under the incongruent condition ($M$ = 913.045 ms, $SD$ = 30.966, $p$ <.001, 95 % CI [849.614, 976.477]). The main effect of emotional valence was not significant and the interaction between two factors was not significant ($p$ =.098, $\eta 2\ p$ =.095), neither.

#### 4.2.2. Accuracy analysis

We analyzed accuracy rates by a repeated measures ANOVA that included the same factors as the RT analysis (see Fig. 3B). The results showed that the main effect of congruency between cue and target was not significant. The main effect of emotional valence was significant, $F$ (1, 28) = 6.375, $p$ =.018, $\eta 2\ p$ =.185, with a significantly higher accuracy rate for negative ($M$ = 0.852, $SD$ = 0.012, 95 % CI [0.827, 0.878]) than for neutral ($M$ = 0.833, $SD$ = 0.014, 95 % CI [0.803, 0.862]). The interaction between congruency and valence was significant, $F$ (1, 28) = 6.613, $p$ =.016, $\eta 2\ p$ =.191. Further simple effects analysis revealed that under the incongruent condition, accuracy for negative valence ($M$ = 0.894, $SD$ = 0.014, 95 % CI [0.866, 0.922]) was significantly higher than for neutral ($M$ = 0.798, $SD$ = 0.026, 95 % CI [0.744, 0.852], $p$ =.006), while there was marginal significant difference between negative ($M$ = 0.810, $SD$ = 0.023, 95 % CI [0.763, 0.858]) and neutral ($M$ = 0.868, $SD$ = 0.015, 95 % CI [0.836, 0.899], $p$ =.058) under the congruent condition.

## 5. General discussion

The current study was performed to assess the modulatory role of task relevance in the influence of emotion on congruency judgments during action understanding. We used a table-tennis action understanding task to activate participants' judgment of congruency between predicted and actual action outcomes. Guided by appraisal theories of emotion (Moors & Fischer, 2019; Scherer & Moors, 2019), which posit that the impact of valence on behaviors is contingent on conscious affective evaluation, we systematically varied the relevance of emotion to the task across three experiments. Consistent with our hypotheses, task relevance altered the impact of emotion on congruency judgments, resulting in distinct manifestation of affective SRC effects (see Table 3).

In experiment 1, emotion served as a task-irrelevant distractor and the incongruent condition resulted in longer reaction times. This finding suggests that the violation of an expected action outcome degrades an individual's action understanding (Zhao et al., 2021). Participants' responses were slower when a stimulus with a negative valence was presented only in the congruent (not in the incongruent) condition. These results are in line with previous findings indicating that emotion as a distractor impairs conflict processing and cognitive control (Dolcos et al., 2013; Dolcos & Denkova, 2014; Dolcos & McCarthy, 2006; Hart et al., 2010). Moreover, the interaction effect indicates that a lateralized affective SRC effect, specifically an incompatible effect, occurred. According to the Theory of Event Coding (Hommel et al., 2001), congruent and incongruent conditions are parallel to neutral and aversive responses, respectively (Vermeylen et al., 2020). The interference of negative distractors appears to make congruent responses more challenging, aligning with our observation. However, we did not observe the facilitation of incongruent responses by negative stimuli, or a compatible effect. This lateralized effect was observed in another study, in which participants made approach and avoidance responses to the (task-irrelevant) gender of emotional faces and reaction times for negative and positive stimuli differed in approach, but not avoidance responses (Welsch et al., 2023). These findings suggest that task relevance modulates the affective SRC effect.

The results of experiment 1 can also be interpreted from the perspective of attention resources. According to the perceptual load theory (Lavie, 1995) and dual competition model (Pessoa, 2009), attention allocation is determined by the perceptual load, with distractors having interfering effects only under low perceptual load conditions. In experiment 1, the target interface displayed emotional stimuli and a table-tennis ball, inevitably generating competition for attentional resources. Under incongruent conditions, a greater perceptual load diverted more attention to congruency judgments, as evidenced by longer response times, leading to less attention to emotional stimuli and comparable behavioral performance with negative and neutral stimuli. By contrast, sufficient attentional resources remained for the emotional distractor under the congruent condition, which had a low perceptual load, resulting

**Table 3**
The Affective Stimulus-Response Compatibility Effects in Three Experiments.

| Experiment 1 (RT) | | Experiment 2 (RT) | | Experiment 3 (ACC) | |
|---|---|---|---|---|---|
| Congruent<br>Negative > Neutral<br>(Incompatible) | Incongruent<br>Negative = Neutral<br>(No effect) | Congruent<br>Negative > Neutral<br>(Incompatible) | Incongruent<br>Negative > Neutral<br>(Contradictory) | Congruent<br>Negative < Neutral<br>(Incompatible) | Incongruent<br>Negative > Neutral<br>(Compatible) |

in different performance with the two types of emotional valence. Similar results have been reported previously. For example, Dou (2021) found that negative faces elicited larger amplitudes of the early visual component N170 only under low load conditions. Shafer (2012) also noted that emotional stimuli caused distraction from a primary task predominantly under low load conditions.

In experiment 2, emotion served as a task-irrelevant target. Participants focused on the spatial information of the stimulus, rather than on valence information. We detected significant main effects of congruency and emotion, but no interaction between them, possibly because the manner in which emotion is presented alters attention allocation. Emotion served as the target stimulus and captured attention in experiment 2, whereas it competed with the target stimuli for attentional resources in experiment 1. However, the captured attention in experiment 2 was limited to the processing of the emotional content rather than orientation information, resulting in equally detrimental effects on performance under the congruent and incongruent conditions.

In addition, the affective SRC effect revealed in experiment 2 needs to be discussed. In previous studies in which emotional stimuli were task irrelevant, such as those in which participants made go and no-go responses to the gender of emotional faces, performance did not differ across valences as they did in valence recognition tasks (Mancini et al., 2022; Mirabella et al., 2023). The design of experiment 2 was similar, as participants processed the spatial features of emotional stimuli to ascertain actual landing points. This spatial focus generated no feature overlap and thus no complete manifestation of the affective SRC effect. This finding is supported by that of Lavender (2007) that the valence-driven affective SRC effect disappeared when participants were required to respond to the spatial information of emotional images. A key contradiction in our findings was the absence of a compatible effect for negative emotions under the incongruent condition, aligning with the finding of Stins et al., (2014) that reaction times are longer for task-irrelevant negative faces than for neutral and positive faces, regardless of movement direction. These results may be attributed to the impairing influence of negative emotions, as evidenced by the demonstration that negative affect impairs cognitive control (Dolcos & Denkova, 2014; Pourtois et al., 2013).

In experiments 1 and 2, participants were not required to identify the emotional content to complete the main tasks, making emotion irrelevant to task performance, and negative emotion had a detrimental effect on congruency judgments during action understanding. In experiment 3, in which emotion served as a task-relevant target, participants had to recognize the emotional valence before obtaining its corresponding actual outcome and completing the main task. Accuracy was lower for negative than for neutral stimuli under congruent conditions ($p$ =.058), but higher under incongruent conditions ($p$ =.006). This trend was also evident for reaction times. Consistent with our hypothesis, the impact of negative emotion on congruency judgments shifted from inhibition to facilitation as the degree of emotion task relevance increased. This finding is in alignment with that of Dou (2021) that participants recognized negative emotions more rapidly under task-relevant conditions.

In addition, the observed interaction revealed a bilateral affective SRC effect when emotions were task relevant, manifesting as incompatible (negative emotions–congruent responses) and compatible (negative emotions–incongruent responses). Similar research in which the task relevance of emotions was manipulated has shown that emotions impact responses only after conscious affective evaluation (Mancini et al., 2020, 2022; Mirabella, 2018; Mirabella et al., 2023). In other words, the effects elicited by emotions may be susceptible to top-down attentional allocation (Dou et al., 2021; Hahn & Gronlund, 2007; Hodsoll et al., 2011; Pessoa et al., 2002; Vromen et al., 2015; Vuilleumier et al., 2001). In Pessoa (2002) study, where bars and emotional faces were presented simultaneously, it was found that amygdala activation by negative stimuli occurred only in tasks focused on identifying face gender, not when the task was to judge bar orientation, highlighting the impact of task-specific attention on emotional processing. In experiment 3 of the present study, the participants were informed in advance that conscious affective evaluation was required to complete the main task. This top-down attentional allocation led to feature overlap between the emotion stimuli and congruency responses, thereby creating compatible and incompatible effects.

Considering the results of our three experiments, we propose that the varied impact of emotion on congruency judgments and the manifestation of the affective SRC effect likely result from combined influences from conscious affective evaluation and top-down attentional allocation. Specifically, participants' top-down attentional resources were allocated preferentially to the target object (i.e., the ball) with the lack of conscious affective evaluation in experiment 1. In accordance with the perceptual load theory (Lavie, 1995), the distracting effect of emotion was observed only under a low perceptual load (i.e., the congruent condition). In experiment 2, emotion was processed in a bottom-up manner and captured substantial attention, which was not beneficial for main task completion. In experiment 3, the task-relevant emotional target conformed to the top-down attentional setting in conjunction with its stimulus characteristics, and captured attentional resources that were deployed for congruency judgments. Thus, emotional stimuli can capture attentional resources and enhance performance in action understanding when they are task relevant; when they are task irrelevant, attentional resources are diverted to emotion processing, reducing performance in the primary task.

### 5.1. Constraints on generality

This study has several limitations. As only congruency judgments in action understanding were examined, whether the modulatory effect of emotion task relevance is generalizable to other types of task needs to be further investigated. In addition, the emotional images in each experimental block were assigned exclusively to congruent or incongruent trials. This lack of counterbalanced in the design could potentially lead to participants recognizing patterns in the appearance of specific emotional images. Additionally, although the participants lacked experience in table tennis, table-tennis stimuli may be overly familiar and potentially enhance arousal in Chinese individuals, raising questions about the generalizability of our findings.

### 5.2. Conclusions

In conclusion, the present study showed that emotions impact congruency judgments during action understanding, and that this effect differed according to the task relevance of emotion. When emotion was task irrelevant, distractor and target objects weakened task performance; when it was task relevant, a facilitative effect was observed. These results support the notion that the impacts of emotions on behavior are contingent on top-down attention allocation.

## CRediT authorship contribution statement

**Yiheng Chen:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Qiwei Zhao:** Visualization, Methodology. **Yueyi Ding:** Investigation. **Yingzhi Lu:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The data, processing code, stimulus materials and experimental procedures are publicly available at https://osf.io/r6vqz/.

## Acknowledgement

This work was supported by grants from National Natural Science Foundation of China (32071089).

## References

Alexander, W. H., & Brown, J. W. (2011). Medial prefrontal cortex as an action-outcome predictor. *Nature Neuroscience, 14*(10), 1338–1344. https://doi.org/10.1038/nn.2921

Alexander, W. H., & Brown, J. W. (2017). The role of the anterior cingulate cortex in prediction error and signaling surprise. *Topics in Cognitive Science, 11*(1), 119–135. https://doi.org/10.1111/tops.12307

Braem, S., King, J. A., Korb, F. M., Krebs, R. M., Notebaert, W., & Egner, T. (2017). The role of ACC in the affective evaluation of conflict. *Journal of Cognitive Neuroscience, 29*(1), 137–149. https://doi.org/10.1162/jocn

Brown, J. W. (2013). Beyond conflict monitoring: Cognitive control and the neural basis of thinking before you act. *Current Directions in Psychological Science, 22*(3), 179–185. https://doi.org/10.1177/0963721412470685

Bush, G., Luu, P., & Posner, M. I. (2000). Cognitive and emotional influences in anterior cingulate cortex. *Trends in Cognitive Sciences, 4*(6), 215–222. https://doi.org/10.1016/S1364-6613(00)01483-2

Chen, M. (1999). Consequences of automatic evaluation: Immediate behavioral predispositions to approach or avoid the stimulus. *Personality and Social Psychology Bulletin, 25*(2), 215–224.

Dignath, D., Eder, A. B., Steinhauser, M., & Kiesel, A. (2020). Conflict monitoring and the affective-signaling hypothesis—An integrative review. *Psychonomic Bulletin and Review, 27*(2), 193–216. https://doi.org/10.3758/s13423-019-01668-9

Dolcos, F., & Denkova, E. (2014). Current emotion research in cognitive neuroscience: Linking enhancing and impairing effects of emotion on cognition. *Emotion Review, 6*(4), 362–375. https://doi.org/10.1177/1754073914536449

Dolcos, F., Iordan, A. D., Kragel, J., Stokes, J., Campbell, R., McCarthy, G., & Cabeza, R. (2013). Neural correlates of opposing effects of emotional distraction on working memory and episodic memory: An event-related fMRI investigation. *Frontiers in Psychology, 4*(JUN), 1–16. https://doi.org/10.3389/fpsyg.2013.00293

Dolcos, F., & McCarthy, G. (2006). Brain systems mediating cognitive interference by emotional distraction. *Journal of Neuroscience, 26*(7), 2072–2079. https://doi.org/10.1523/JNEUROSCI.5042-05.2006

Dou, H., Liang, L., Ma, J., Lu, J., Zhang, W., & Li, Y. (2021). Irrelevant task suppresses the N170 of automatic attention allocation to fearful faces. *Scientific Reports, 11*(1), 1–10. https://doi.org/10.1038/s41598-021-91237-9

Dreisbach, G., & Fischer, R. (2015). Conflicts as aversive signals for control adaptation. *Current Directions in Psychological Science, 24*(4), 255–260. https://doi.org/10.1177/0963721415569569

Eder, A. B., & Rothermund, K. (2008). When do Motor behaviors (mis)match affective stimuli? an evaluative coding view of approach and avoidance reactions. *Journal of Experimental Psychology: General, 137*(2), 262–281. https://doi.org/10.1037/0096-3445.137.2.262

Etkin, A., Egner, T., & Kalisch, R. (2011). Emotional processing in anterior cingulate and medial prefrontal cortex. *Trends in Cognitive Sciences, 15*(2), 85–93. https://doi.org/10.1016/j.tics.2010.11.004

Faul, F., Erdfelder, E., Lang, A.-G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods, 39*(2), 175–191.

Hahn, S., & Gronlund, S. D. (2007). Top-down guidance in visual search. *Psychonomic Bulletin & Review, 14*(1), 159–165.

Hart, S. J., Green, S. R., Casp, M., & Belger, A. (2010). Emotional priming effects during stroop task performance. *NeuroImage, 49*(3), 2662–2670. https://doi.org/10.1016/j.neuroimage.2009.10.076

Hodsoll, S., Viding, E., & Lavie, N. (2011). Attentional capture by irrelevant emotional distractor faces. *Emotion, 11*(2), 346–353. https://doi.org/10.1037/a0022771

Hommel, B., Müsseler, J., Aschersleben, G., & Prinz, W. (2001). The theory of event coding (TEC): A framework for perception and action planning. *Behavioral and Brain Sciences, 24*(5), 849–878. https://doi.org/10.1017/S0140525X01000103

Inzlicht, M., Bartholow, B. D., & Hirsh, J. B. (2015). Emotional foundations of cognitive control. *Trends in Cognitive Sciences, 19*(3), 126–132. https://doi.org/10.1016/j.tics.2015.01.004

Kanske, P., & Kotz, S. A. (2011a). Emotion speeds up conflict resolution: A new role for the ventral anterior cingulate cortex? *Cerebral Cortex, 21*(4), 911–919. https://doi.org/10.1093/cercor/bhq157

Kanske, P., & Kotz, S. A. (2011b). Emotion triggers executive attention: Anterior cingulate cortex and amygdala responses to emotional words in a conflict task. *Human Brain Mapping, 32*(2), 198–208. https://doi.org/10.1002/hbm.21012

Klauer, K. C., & Musch, J. (2002). Goal-dependent and goal-independent effects of irrelevant evaluations. *Personality and Social Psychology Bulletin, 28*(6), 802–814. https://doi.org/10.1177/0146167202289009

Kozlik, J., Neumann, R., & Lozo, L. (2015). Contrasting motivational orientation and evaluative coding accounts: On the need to differentiate the effectors of approach/avoidance responses. *Frontiers in Psychology, 6*(MAY), 1–10. https://doi.org/10.3389/fpsyg.2015.00563

Krieglmeyer, R., & Deutsch, R. (2010). Comparing measures of approach-avoidance behaviour: The manikin task vs. two versions of the joystick task. *Cognition and Emotion, 24*(5), 810–828. https://doi.org/10.1080/02699930903047298

Lavender, T., & Hommel, B. (2007). Affect and action: Towards an event-coding account. *Cognition and Emotion, 21*(6), 1270–1296. https://doi.org/10.1080/02699930701438152

Lavie, N. (1995). Perceptual load as a Necessary condition for selective attention. *Journal of Experimental Psychology: Human Perception and Performance, 21*(3), 451–468. https://doi.org/10.1037/0096-1523.21.3.451

Lu, Y., Yang, T., Hatfield, B. D., Cong, F., & Zhou, C. (2020). Influence of cognitive-motor expertise on brain dynamics of anticipatory-based outcome processing. *Psychophysiology, 57*(2), 1–13. https://doi.org/10.1111/psyp.13477

Mancini, C., Falciati, L., Maioli, C., & Mirabella, G. (2020). Threatening facial expressions impact goal-directed actions only if task-relevant. *Brain Sciences, 10*(11), 1–18. https://doi.org/10.3390/brainsci10110794

Mancini, C., Falciati, L., Maioli, C., & Mirabella, G. (2022). Happy facial expressions impair inhibitory control with respect to Fearful facial expressions but only when task-relevant. *Emotion, 22*(1), 142–152. https://doi.org/10.1037/emo0001058

Mirabella, G. (2018). The weight of emotions in decision-making: How fearful and happy facial stimuli modulate action readiness of goal-directed actions. *Frontiers in Psychology, 9*(AUG), 1–8. https://doi.org/10.3389/fpsyg.2018.01334

Mirabella, G., Grassi, M., Mezzarobba, S., & Bernardis, P. (2023). Angry and happy expressions affect Forward gait initiation only when task relevant. *Emotion, 23*(2), 387–399. https://doi.org/10.1037/emo0001112

Moors, A., & Fischer, M. (2019). Demystifying the role of emotion in behaviour: Toward a goal-directed account. *Cognition and Emotion, 33*(1), 94–100. https://doi.org/10.1080/02699931.2018.1510381

Moreau, Q., Candidi, M., Era, V., Tieri, G., & Aglioti, S. M. (2020). Midline frontal and occipito-temporal activity during error monitoring in dyadic motor interactions. *Cortex, 127*, 131–149. https://doi.org/10.1016/j.cortex.2020.01.020

Moreau, Q., Tieri, G., Era, V., Aglioti, S. M., & Candidi, M. (2022). The performance monitoring system is attuned to others' actions during dyadic motor interactions. *Cerebral Cortex (New York, N.Y. : 1991), 33*(1), 222–234. https://doi.org/10.1093/cercor/bhac063.

Pessoa, L. (2009). How do emotion and motivation direct executive control? *Trends in Cognitive Sciences, 13*(4), 160–166. https://doi.org/10.1016/j.tics.2009.01.006

Pessoa, L., McKenna, M., Gutierrez, E., & Ungerleider, L. G. (2002). Neural processing of emotional faces requires attention. *Proceedings of the National Academy of Sciences of the United States of America, 99*(17), 11458–11463. https://doi.org/10.1073/pnas.172403899

Pourtois, G., Schettino, A., & Vuilleumier, P. (2013). Brain mechanisms for emotional influences on perception and attention: What is magic and what is not. *Biological Psychology, 92*(3), 492–512. https://doi.org/10.1016/j.biopsycho.2012.02.007

Roh, D., Chang, J. G., & Kim, C. H. (2016). Emotional interference modulates performance monitoring in patients with obsessive-compulsive disorder. *Progress in Neuro-Psychopharmacology and Biological Psychiatry, 68*, 44–51. https://doi.org/10.1016/j.pnpbp.2016.03.006

Rotteveel, M., & Phaf, R. H. (2004). Automatic affective evaluation does not automatically predispose for arm flexion and extension. *Emotion, 4*(2), 156–172. https://doi.org/10.1037/1528-3542.4.2.156

Scherer, K. R., & Moors, A. (2019). The emotion process: Event appraisal and component differentiation. *Annual Review of Psychology, 70*(August), 719–745. https://doi.org/10.1146/annurev-psych-122216-011854

Shackman, A. J., Salomons, T. V., Slagter, H. A., Fox, A. S., Winter, J. J., & Davidson, R. J. (2011). The integration of negative affect, pain and cognitive control in the cingulate cortex. *Nature Reviews Neuroscience, 12*(3), 154–167. https://doi.org/10.1038/nrn2994

Shafer, A. T., Matveychuk, D., Penney, T., O'Hare, A. J., Stokes, J., & Dolcos, F. (2012). Processing of emotional distraction is both automatic and modulated by attention: Evidence from an event-related fMRI investigation. *Journal of Cognitive Neuroscience, 24*(5), 1233–1252. https://doi.org/10.1162/jocn_a_00206

Stins, J. F., Lobel, A., Roelofs, K., & Beek, P. J. (2014). Social embodiment in directional stepping behavior. *Cognitive Processing, 15*(3), 245–252. https://doi.org/10.1007/s10339-013-0593-x

Vermeylen, L., Wisniewski, D., González-García, C., Hoofs, V., Notebaert, W., & Braem, S. (2020). Shared neural representations of cognitive conflict and negative affect in the medial frontal cortex. *Journal of Neuroscience, 40*(45), 8715–8725. https://doi.org/10.1523/JNEUROSCI.1744-20.2020

Vromen, J. M. G., Lipp, O. V., & Remington, R. W. (2015). The spider does not always win the fight for attention: Disengagement from threat is modulated by goal set. *Cognition and Emotion, 29*(7), 1185–1196. https://doi.org/10.1080/02699931.2014.969198

Vuilleumier, P., Armony, J. L., Driver, J., & Dolan, R. J. (2001). Effects of attention and emotion on face processing in the human brain. *Neuron, 30*, 829–841.

Welsch, R., Hecht, H., & Stins, J. (2023). Task-relevant social cues affect whole-body approach-avoidance behavior. *Scientific Reports, 13*(1), 1–13. https://doi.org/10.1038/s41598-023-35033-7

Xia, X., Wang, D., Song, Y., Zhu, M., Li, Y., Chen, R., & Zhang, J. (2021). Involvement of the primary motor cortex in the early processing stage of the affective stimulus-response compatibility effect in a manikin task. *NeuroImage*, *225*(June 2020). https://doi.org/10.1016/j.neuroimage.2020.117485.

Xue, S., Cui, J., Wang, K., Zhang, S., Qiu, J., & Luo, Y. (2013). Positive emotion modulates cognitive control: An event-related potentials study. *Scandinavian Journal of Psychology, 54*(2), 82–88. https://doi.org/10.1111/sjop.12031

Yuan, J., Xu, S., Yang, J., Liu, Q., Chen, A., Zhu, L., Chen, J., & Li, H. (2011). Pleasant mood intensifies brain processing of cognitive control: ERP correlates. *Biological Psychology, 87*(1), 17–24. https://doi.org/10.1016/j.biopsycho.2011.01.004

Zhao, Q., Lu, Y., Jaquess, K. J., & Zhou, C. (2018). Utilization of cues in action anticipation in table tennis players. *Journal of Sports Sciences, 36*(23), 2699–2705. https://doi.org/10.1080/02640414.2018.1462545

Zhao, Q., Wang, Y., Chen, Y., Wang, Y., Zhou, C., & Lu, Y. (2021). Expertise influences congruency monitoring during action observation at the motor level. *Social Cognitive and Affective Neuroscience, 16*(12), 1288–1298. https://doi.org/10.1093/scan/nsab078